

Grafting embedded poly(ionic liquid) brushes on biomimetic sharklet resin surface for anti-biofouling applications

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ABSTRACT

In this work, we combined 3D printing technology and subsurface-initiated ring opening metathesis polymerization (ROMP) to construct a multi-scale brush-based antifouling surface. A biomimetic sharklet structured substrate embedded with ROMP initiator was prepared by 3D printing with commercial acrylic resin. Then, the embedded poly(ionic liquid) brushes were grafted onto the as-prepared biomimetic sharklet via subsurface-initiated ROMP of a novel ionic liquid monomer containing benzotriazole and imidazole groups ([BNIm][Br]). A series of characterization including infrared spectroscopy, X-ray photoelectron spectroscopy and atomic force microscope proved the successful grafting of poly([BNIm][Br]) brushes on the as-prepared surface. X-ray photoelectron spectroscopy etching and friction tests demonstrated that poly([BNIm][Br]) can be grafted not only on the surface but on the subsurface of the as-prepared surface, and the as-prepared poly(ionic liquid) brush-based surfaces showed satisfactory wear resistance compared to traditional surface initiated ROMP. Subsequently, we evaluated the anti-biofouling properties of poly([BNIm][Br]) brushes. The results indicated that poly(ionic liquid) brushes can obviously resist adhesion of bovine serum albumin and have good anti-bacterial activity against both *E. coli* and *S. aureus*. The as-prepared poly(ionic liquid) brush-based biomimetic surface also exhibited considerable antifouling performance for microalgae (*Porphyridium* and *Dunaliella*) due to the synergistic effect of the surface composition and microstructure.

1. Introduction

Marine biofouling is a general term used to describe undesired marine animals, plants, and microorganisms that grow on the surface of various means of aquatic transport, seabed, and other marine equipment, which seriously restricts the development and utilization of marine resources. [1,2] It is responsible for increasing the navigation resistance of ships and the amount of fuel consumed, reducing the speed of ships, causing anaerobic corrosion of metals, etc [3]. According to statistics, the economic losses caused by biofouling exceed US\$150 billion in terms of transportation costs every year [4]. At present, antifouling coating is still the most widespread measure used in marine antifouling. After Tributyltin (TBT) was phased out of use from January 1st, 2008 by the International Maritime Organization (IMO) because of its biotoxicity, organic copper and zinc based antifouling coatings were widely developed. [5] Unfortunately, these antifouling agents continue

to remain harmful to the marine environment and potentially threaten human health [6]. Hence, there is an urgent need to develop new varieties of environmentally friendly antifouling materials.

Polymer brushes are polymer film structures consisting of relatively high density polymer chains end-tethered to a surface, [7] which can endow the surface of the material with a variety of useful properties, such as biocompatibility, anti-corrosion, antifouling, molecular recognition, etc. [8] In recent years, polymer brushes have developed rapidly in the field of antifouling; not only have the preparation methods improved, but the diversity of antifouling monomers has also broadened [9,10]. Recently, Zhou et al. prepared embedded polymer brushes having thicknesses ranging from nanometer to micron scale via subsurface-initiated ATRP [11]. The hydrolytic stability and antifouling performance of the as-prepared 3-sulfopropyl methacrylate potassium salt (SPMA) polymer brushes were improved significantly by increasing the grafting thickness [12]. In addition, more varieties of polymers that

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exhibit antifouling effect have been developed, including polyglycol (PEG) and its derivatives [13], poly(2-alkyl-2-oxazine) [14], zwitter-ionic polymers [15], cationic polymer-containing quaternary ammonium salt [16], and poly(ionic liquid)s [17].

As always, considerable attention has been paid to the cationic polymer due to its satisfactory sterilization capability [18]. Studies have shown that when a bacterium comes in contact with the cationic polymer side chains such as quaternary ammonium and imidazolium cations, these cationic units attach to the bacterium and penetrate the cell membrane, causing cell death [19]. Many ionic liquids (ILs) contain imidazole, pyridinium, and ammonium groups, and have been shown to inhibit growth of bacteria and fungi in a manner similar to cationic antimicrobials [20]. Zhao et al. synthesized a bromide ionic liquid (IL (Br)) on the membrane surface and modified it using Michael addition [21]. The IL(Br) substrate can effectively kill *S. aureus* and *E. coli* due to the synergistic effects of quaternary ammonium groups, long chain alkanes, and imidazole rings. Poly(ionic liquid)s (PILs) are an emerging group of functional polymers containing backbone and ionic liquid in monomer-repeating moieties [22]. From monomers to polymer, certain special properties of ionic liquid, such as thermal stability, high ionic conductivity, and antibacterial property, are transferred to PILs [23]. The Zhou Research Group have prepared high-density poly(ionic liquid) brushes via surface-initiated ROMP and evaluated their antifouling properties. Poly(ionic liquid) brushes based on imidazolium salts can obviously resist adhesion of spores, and the counter-anions play a crucial role in their antifouling performance [24]. Du et al. have prepared poly(vinylidene fluoride) blended membranes via block copolymers of PILs, and the as-prepared membranes showed high rejection and good antifouling property with increase in the chain length of PILs, owing to the high charge density and strong electrostatic repulsion [25].

Besides the influence of chemical composition, surface microstructure also plays an important role in antifouling. It has been reported that the sharklet structure can effectively reduce the attachment of pollutants [26,27]. Herein, the biomimetic sharklet structure is prepared via Digital Light Processing (DLP, a kind of 3D printing photocuring molding) of initiator-embedded photosensitive resin [28]. After that, the novel poly(ionic liquid) brushes (p [BNIm][Br]) were grafted on the as-prepared surface via subsurface-initiated ROMP. The poly(ionic liquid) compositions and the antifouling performance, including protein adsorption, algal adhesion, and anti-bacterial qualities, of the poly(ionic liquid) coated surfaces were examined, and the as-prepared p[BNIm][Br]-based surface exhibited obvious antifouling and antimicrobial properties.

2. Experimental section

2.1. Materials

Grubbs 2nd generation catalyst (98 %), imidazole (98 %), benzotriazole (98 %), and dicyclopentadiene (97 %) were used as received from Sigma-Aldrich (Merck KGaA, Germany). 5-norbornene-2-carboxylic acid (97 %, mixture of endo and exo) was purchased from Alfa. Norbornylmethylene bromide and ROMP initiator (BNC) was synthesized through the reported methods [12]. Sodium hydroxide, dimethyl sulfoxide (DMSO), isopropanol, and *n*-hexane were obtained from Chemical Reagent Company of Shanghai (Shanghai, China). The commercial photosensitive acrylic resin was purchased from Chuangxiang 3D Ltd. (Shenzhen, China).

2.2. Synthesis of ionic liquid monomer 1-(1H-Benzotriazolyl) methyl-3-norbornene methyl-1H-imidazolium Bromide ([BNIm][Br])

As shown in Scheme S1, the intermediate **1a**, 1-(chloromethyl)-1H-benzotriazole, was synthesized according to procedure given in previously reported literature [29]. Then, 1.36 g (20 mmol) imidazole, 1.6 g (40 mmol) sodium hydroxide, and 10 mL DMSO were added to the 50

mL round-bottom flask. After dissolving, the mixture was heated to 60 °C for 1 h. Subsequently, 3.34 g (20 mmol) of 1-(chloromethyl)-1H-benzotriazole was dripped in slowly, and the color of the mixture changed rapidly from light yellow to brown, and then gradually to orange. After stirring at 60 °C for 18 h, the reaction solution was transferred to a 250 mL beaker and 150 mL deionized water was added. The mixture was kept standing overnight and the precipitation was filtered and washed with deionized water to obtain the intermediate **2a**, 1-(imidazolylmethyl)-1H-benzotriazole (pale yellow needle solid).

Next, 1.99 g (10 mmol) of intermediate **2a** and 3.74 g (20 mmol) of norbornylmethylene bromide were dissolved with 20 mL of isopropanol in a 50 mL two-neck round-bottom flask. The mixture was heated to 80 °C and refluxed for 48 h. After the reaction was complete, the organic solvent was evaporated under reduced pressure and the supernatant oily liquid (excess norbornylmethylene bromide) was then removed. The underlying brown viscous liquid ([BNIm][Br]) was washed with *n*-hexane and then dried in a vacuum oven, with a total yield of 50 % and stored at ~ 4 °C.

¹H-NMR (600 MHz, CDCl₃), (ppm): 8.80 (m, 1 H), 8.04 (d, 1 H), 7.83 (d, 1 H), 7.80 (m, 2 H), 7.64 (t, 1 H), 7.43 (t, 1 H), 6.79 (s, 2 H), 6.30 (m, 1 H), 6.13 (m, 1 H), 4.26 (m, 1 H), 3.93 (m, 1 H), 2.89 (s, 1 H), 2.64 (m, 1 H), 1.99 (m, 1 H), 1.39 (d, 2 H), 1.26 (d, 2 H),

¹³C-NMR (150 MHz, CDCl₃), (ppm): 145.95, 139.65, 137.87, 135.49, 132.66, 129.72, 125.34, 122.22, 122.16, 119.86, 111.41, 58.22, 54.72, 49.65, 44.21, 42.49, 30.17, 19.15;

High-resolution mass spectrum: calcd [M+H]⁺, 386.0975, found, 306.1728 (Bromide ion was not found).

2.3. 3D printing of initiator-containing resin with different surface morphologies

To prepare the initiator-containing resin, ROMP initiator (3-butenyl-5-norbornene-2-carboxylate, BNC) was added to the resin base of the 3D printer. After degassing for 15 min, the photosensitive acrylic resin containing 2 wt % initiator BNC was ready for printing. The process of as-prepared resin photo-curing is similar to the procedure reported in literature [30].

Briefly, a solid 3D model was designed by CAD software and was firstly digitally sliced into a series of 2D layers. During printing, the exposure time for each layer was 3 s, and the lift height was 25 μm. Once finished, the cured resin was ultrasonically rinsed in ethanol solution to remove the absorbed small molecular monomers, and then post-cured under UV light for 5 min to further cross-link the resin surface. As shown in Fig. 1b, biomimetic sharklet surface were referred to the previous reports [31,32]. To imitate the spine structure of the shark skin surface, regular ribs of different lengths were constructed.

2.4. Subsurface-initiated ring-opening metathesis polymerization

The polymerization process was carried out under argon. As shown in Fig. 1a, the subsurface-initiated ROMP was performed as follows: 10 mg of Grubbs 2nd generation catalyst was dissolved in 5 mL anhydrous DMF. The initiator-containing resin surface was placed in the catalyst-containing solution (2 mg/mL). At the same time, 500 mg of [BNIm][Br] was fully dissolved in 3 mL anhydrous DMF. After 15 min, the [Ru]-functionalized resin was taken out, and was carefully washed with anhydrous DMF to remove the absorbed catalyst. Then, the resin was transferred into the as-prepared solution of [BNIm][Br] (0.43 M) and kept for 40 min. Finally, the modified resin was taken out and cleaned with DMF and stored in a drying cabinet for further study. The [BNIm][Br] modified resin via traditional SI-ROMP were also prepared. In contrast, the initiator was anchored to the cured resin surface by pyrocatechol, instead of copolymerizing with the resin prepolymer. The time of polymerization and the concentration of catalyst and [BNIm][Br] were the same as subsurface-initiated polymerization [33].

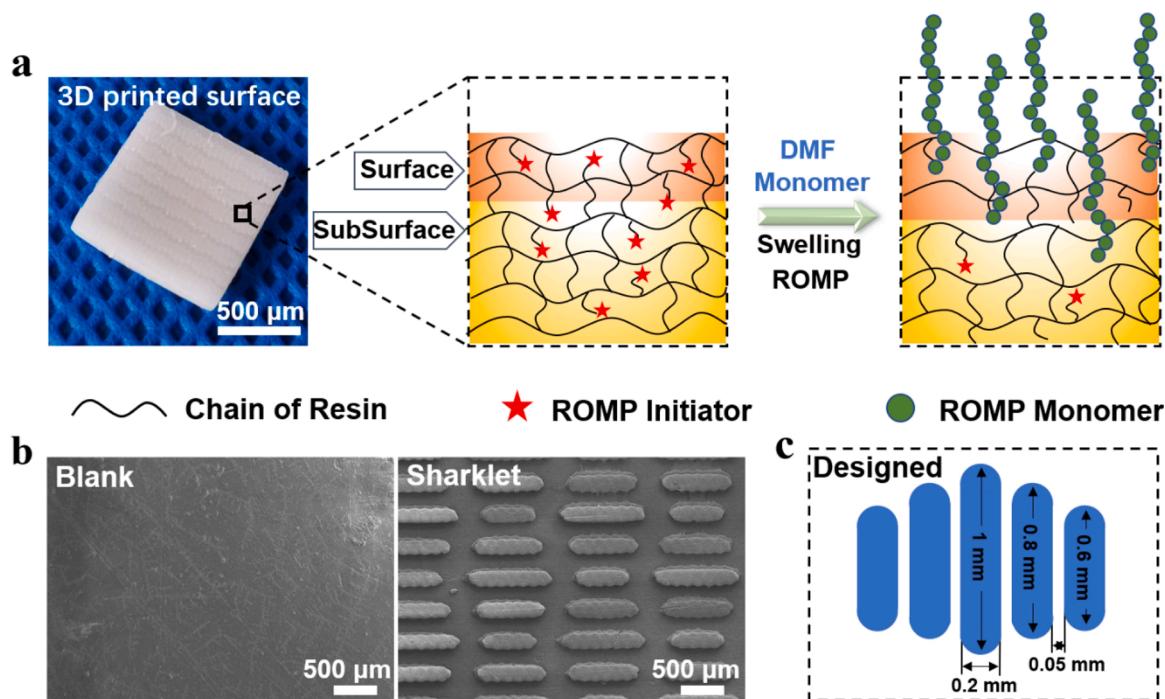


Fig. 1. (a) Schematic illustration of polymer brush-modified 3D printed resin surface via subsurface-initiated ROMP; (b) surface topography and (c) designed dimensions of 3D printed resin.

2.5. Characterization

$^1\text{H-NMR}$ and $^{13}\text{C-NMR}$ spectra of the $[\text{BNIm}][\text{Br}]$ were recorded on a 600 MHz spectrometer (Bruker 600 MHz). High-resolution mass spectrum was tested on Bruker MAXIS spectrometer. The chemical composition information of the surfaces was obtained by X-ray photoelectron spectroscopy (XPS) spectra and employing $\text{Al K}\alpha$ excitation radiation. The binding energy of C1 s at 284.8 eV from adventitious carbon was taken as a reference. ATR-FTIR spectrum were tested by Bruker Tensor II infrared spectrometer. The hydrophilicity of the resin surfaces was measured by DSA-100 optical contact angle meter (Kruss Company, Ltd, Germany) at room temperature. The surface morphology before and after polymer brushes modified was performed using a Multimode AFM (Asylum Research MFP-3D Origin, Oxford instruments) in the tapping mode. The thickness of polymer layers was measured with a spectroscopic ellipsometer (Gaertner model L116). Each friction test is reciprocated at a sliding speed of 2 mm /s under a load of 0.5 N on a UMT-Tribo friction and wear tester (Bruker). The upper friction pair is steel ball (diameter: 6.8 mm), the lower friction pair is sample with the “Smooth” surface, and the lubricating medium is aqueous solution of CTAB with a concentration of 5 mmol /L.

The protein adsorption assay on the surface was determined using a quartz crystal microbalance with dissipation measurement (QCM-D) at 25 °C. The gold-plated quartz chip QSX-301 (QSense) was used in this test and the quality of the adhesion was obtained by the following formula: [34]

$$\Delta m = -C \frac{\Delta f}{n} \quad (1)$$

The Live/Dead BacLight method was used to carry out the quantitative test of the two bacteria—*E. coli* and *S. aureus*. Attached live bacteria (green dots) and dead bacteria (red dots) were observed and counted using a fluorescence microscope and ImageJ software. Two typical types of microalgae—*Dunaliella* and *Chlorella*—were purchased from the Freshwater Algae Culture Collection at the Institute of Hydrobiology (FACHB). The static attachment assays were performed as per methods reported in literature [12]. The HAAKE RS6000 rheometer

was used to create a dynamic environment and a plate-and-plate model was employed in this experiment [35]. As shown in Fig. S2, the distance between the clamp and test surface was 1 cm and the distilled water was injected into the gap. The test sample, which has been adhered by the microalgae after static test, was stuck on the lower plate. The rotating speed of the clamp was 500 r/min and the duration time was 1 min. All samples were tested under the same conditions to ensure the consistent shear stress. After the test, the statistical method of microalgae adhesion is the same as the static experiment.

3. Results and discussion

3.1. Surface characterization

The successful polymerization of $\text{p}[\text{BNIm}][\text{Br}]$ was first confirmed by Fourier-transform infrared spectroscopy (FT-IR) and X-ray photoelectron spectroscopy (XPS) spectrum. In order to measure the positions of functional groups of polymer brushes, the $[\text{BNIm}][\text{Br}]$ monomer was analyzed using FT-IR. The representative and strong absorption peaks of $[\text{BNIm}][\text{Br}]$ are indicated in Fig. 2a. The characteristic absorption peaks located at 3107 cm^{-1} , 744 cm^{-1} , and 625 cm^{-1} , can be attributed to the skeleton vibration [36] and C–H out-of-plane bending vibration peak on the benzene ring of benzotriazole [37]. Furthermore, the typical vibration bands at 1448 cm^{-1} and 1160 cm^{-1} were attributed to the imidazolium ring cation [24]. On this basis, the infrared spectra of the resin surface before and after modification were compared in Fig. 2b. Although some absorption peaks from the vibration of $[\text{BNIm}][\text{Br}]$ are covered by the resin, it can be clearly seen that new characteristic absorption peaks (3110 cm^{-1} , 1447 cm^{-1} , 1156 cm^{-1} , and 744 cm^{-1}) appear on the modified surface, which are almost the same as those of the $[\text{BNIm}][\text{Br}]$ in Fig. 2a. It indicates that $\text{p}[\text{BNIm}][\text{Br}]$ successfully graft onto resin surface. Fig. 2c displays the XPS full survey spectra of the unmodified surface and $\text{p}[\text{BNIm}][\text{Br}]$ modified surface. C, O, and Si signals were found on both two surfaces, which are elements derived from the photosensitive acrylic resin itself. After modification by $\text{p}[\text{BNIm}][\text{Br}]$, it can be found that the signal of N element increases significantly. Moreover, the N 1s spectrum of the modified resin surface

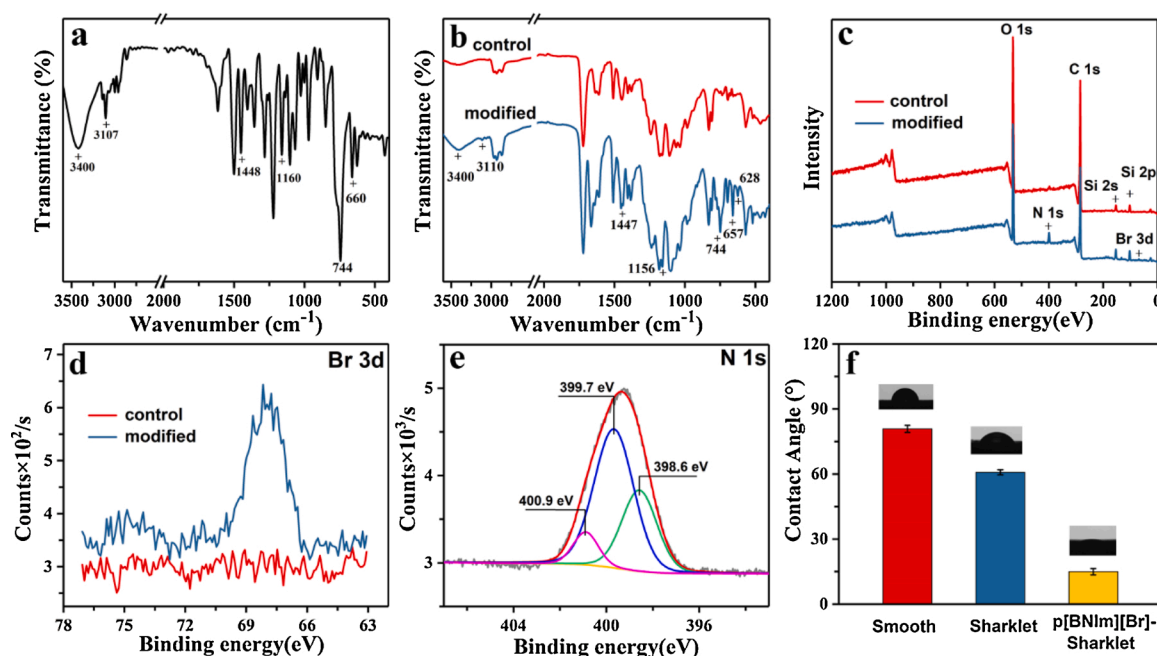


Fig. 2. (a) FT-IR spectra of [BNIm][Br] monomer; (b) FT-IR spectra of the resin surface before and after p[BNIm][Br] modification; (c) XPS survey spectra and (d) Br 3D spectra of blank resin and p[BNIm][Br] modified resin; (e) Curve-fitted XPS spectra of N 1s of the modified resin; (f) The contact angles of water droplets on different surfaces.

shows three peaks located at 398.6, 399.7, and 400.9 eV, which correspond to the C–N, CN=N=N bonds in benzotriazole, imidazole ring and ammonium nitrogen of imidazolium cation [37,38]. Additional support for the successful modification of the p[BNIm][Br] comes from the appearance of Br 3D signal (see Fig. 2d). As shown in Fig. 2f, the hydrophilicity of the resin surfaces was measured from contact angle (CA) measurements. The water contact angles of the unmodified “Smooth” and “Sharklet” surfaces are $81^\circ \pm 1.6^\circ$ and $61^\circ \pm 1.1^\circ$, respectively; the change of surface wettability results from the surface microstructure of the resin. After the “Sharklet” surface was modified by p[BNIm][Br], the CA rapidly decreased to $15^\circ \pm 1.5^\circ$, which is due to the excellent hydrophilicity of the grafting polymer brushes.

The surface morphology of the polymer brushes was observed with

scanning electron microscopy (SEM) and atomic force microscope (AFM) ($1 \mu\text{m} \times 1 \mu\text{m}$). As shown in Fig. 3a, b, and Fig. S2, the unmodified resin surface appears relatively smooth with root-mean-square (RMS) roughness of 7.22 nm. After modification, the small grains are seen uniformly distributed over the whole surface, and the roughness observed as significantly increased to 11.49 nm, which shows that p[BNIm][Br] brushes have undergone successful grafting [39]. In order to estimate the thickness of polymer brushes, the p[BNIm][Br] are prepared via SI-ROMP on a silicon wafer under the same polymerization procedure [33]. The result of ellipsometry indicates that the average thickness of polymer brushes is about 60 nm.

To compare the difference between subsurface initiated ROMP and surface initiated ROMP (SI-ROMP), the growth depth and strength of the

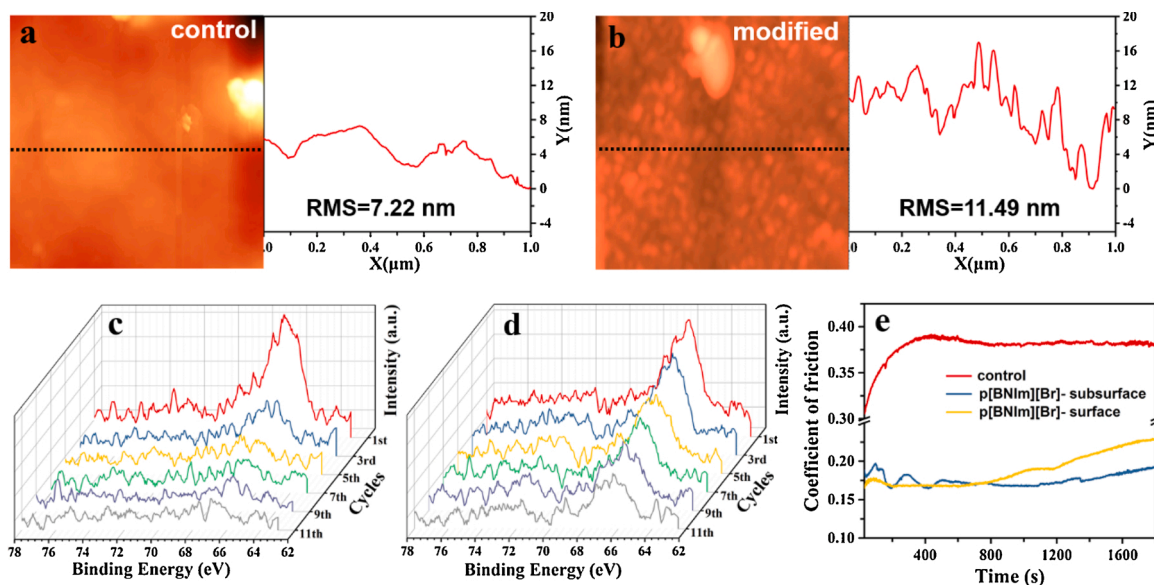


Fig. 3. AFM images and corresponding high profiles of (a) unmodified resin and (b) p[BNIm][Br]-modified resin surface; Br 3d signal of p[BNIm][Br]-modified resin via (c) surface-initiated ROMP and (d) subsurface-initiated ROMP; (e) The friction behavior of different substrates measured using UMT-Tribo.

polymer brushes were tested using XPS argon cluster etching and Universal Mechanical Tester (UMT) friction testing machine [40]. In this study, a 5 keV argon cluster was scattered over a $200\ \mu\text{m} \times 200\ \mu\text{m}$ area to create the depth profile. As shown in Fig. 3c and d, the Br 3d signal (which only exists in the p[BNIIm][Br]) after each etching was tested; here numbers 1–11 represent the etching cycles (120 s each). In Fig. 3d, the Br 3d signal weakens significantly with the increase in sputtering time. After the 5th etching, the Br 3d signal is so weak that it is almost undetectable. However, for p[BNIIm][Br] modified by subsurface initiated ROMP, the Br 3d signal can still be detected even after 11 etching cycles, indicating that the polymer brushes can grow at a deeper depth. This is due to the resin surface swelling in the organic solvent, and some of the initiators embedded in the resin, that is in contact with the monomer solution, leading the polymer brushes to grow from the subsurface. Fig. 3e depicts the water lubrication performance of different resin surfaces. The coefficient of friction (COF) of the unmodified resin surface remains around 0.38, while the COF significantly drops to around 0.17 after the surface was modified by p[BNIIm][Br]. The significant decrease of COF is due to the hydration layer formed by the hydrophilic p[BNIIm][Br] at the friction interface. After 13 min of friction, the COF of the surface begin to rise because the resin matrix was exposed as the modified polymer brushes gradually wore away. Compared to the surface-initiated ROMP, the p[BNIIm][Br]-based surface via subsurface initiated ROMP can maintain a low COF for 22 min, after which the coefficient of friction slowly rises. The result indicates that the as-prepared polymer brushes via subsurface initiated ROMP have better wear resistance, and hints that the polymer brushes can be successfully grafted onto the surface and subsurface of the resin.

3.2. Antifouling performance

The anti-protein adsorption property of p[BNIIm][Br] was tested with QCM-D. As shown in Fig. 4a and b, the BSA solution (20 mg/mL) was pumped into the chamber and washed with ultra-pure water after the

chip surface adsorption stabilized. The amount of protein adsorbed on bare Au and p[BNIIm][Br]-modified Au-coated quartz chip were calculated from the change in frequency. The mass of the adhering layer for Au and p[BNIIm][Br] are 294.7 and 136.4 ng/cm², respectively. The decrease in adsorption reveals that the p[BNIIm][Br] has a certain anti-protein effect, which is due to the highly hydrophilic surface of Au-coated quartz [41].

Benzotriazole (BTA) is an important aromatic heterocyclic compound, which is often used as metal rust and corrosion inhibitor [42,43]. In recent years, studies have shown that benzotriazole derivatives have antibacterial effects [44]. Fig. S3 shows the antibacterial performance of different resin surfaces. It can be clearly seen that a large number of *E. coli* or *S. aureus* attach to the unmodified surface. In order to calculate the attachment density of bacteria on the surface and explore the antibacterial mechanism of p[BNIIm][Br], live/dead bacterial cell staining assays were carried out. As shown in Fig. 4c and Fig. S3, nearly all *S. aureus* can survive on the unmodified “Smooth” surface, with a statistical attachment density of $19,080 \pm 2365/\text{mm}^2$. For the “Sharklet” surface, the density of bacteria attached on the surface drops slightly ($16,363 \pm 1243/\text{mm}^2$) and an exceedingly small number of bacteria died ($112 \pm 73/\text{mm}^2$). After modification by p[BNIIm][Br], the surface exhibits considerable antifouling properties, in that a much lower attachment density and more dead cells were observed ($763 \pm 252/\text{mm}^2$ live and $1002 \pm 266/\text{mm}^2$ dead). The same regularity can be seen in the attachment of *E. coli* groups (see Fig. 4d), in which the “Sharklet” surface only reduces about 15 % attachment of bacteria and p[BNIIm][Br]. The “Sharklet” surface shows excellent anti-attachment ability and bacterial killing property. The bactericidal properties are mainly derived from the benzotriazole group and imidazole cationic groups of the p[BNIIm][Br] side chains [23,44] and the cationic units cause cell death by destroying the membrane integrity [23]. The anti-attachment performance may be due to the good hydrophilic property of poly(ionic liquid) modified “Sharklet” surface, making it difficult for microorganisms to attach to the surface.

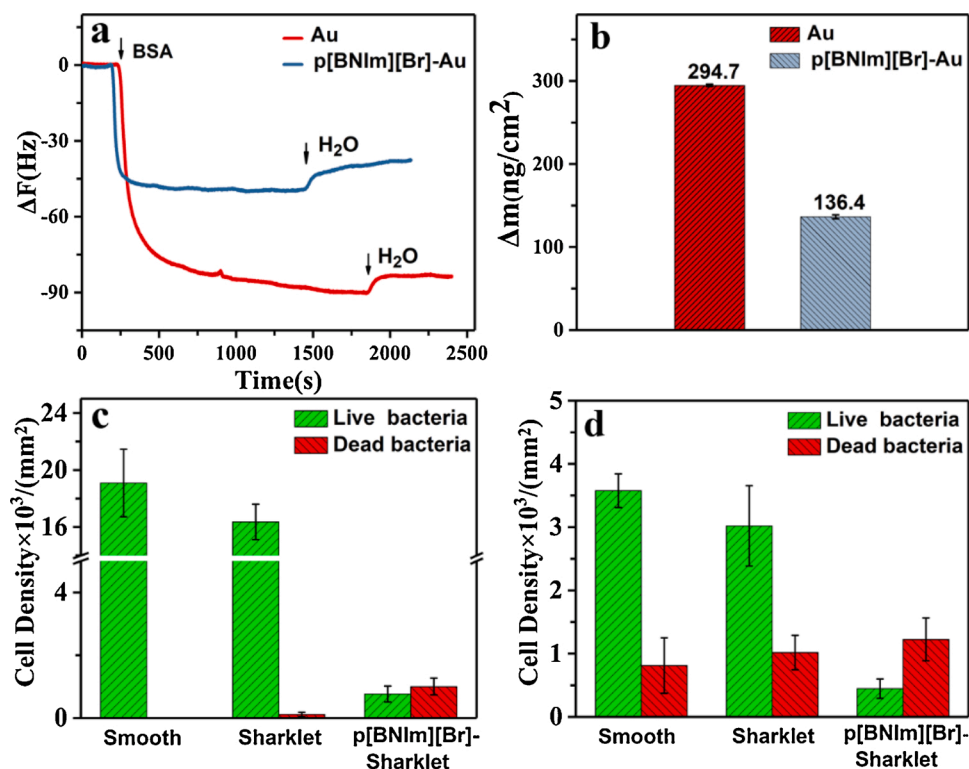


Fig. 4. (a) QCM-D adsorption curves and (b) the corresponding adsorption masses of bare and p[BNIIm][Br]-modified Au chips in the presence of BSA (20 mg/mL) solutions; The statistical results of dead and alive bacteria of (c) *S. aureus* and (d) *E. coli* attached on different surfaces.

To better evaluate the antifouling performance of p[BNIm][Br]-modified surface, static adhesion tests using *Dunaliella* and *Porphyridium* were performed. Three samples of each group were immersed in microalgae culture suspension (reaches 10^7 cells/mL) at 25 °C. After 24 h culture, the resin was taken out and washed slightly using artificial seawater. Fluorescence microscope was used to randomly select 20 fields for observation of each group of samples. Because the microstructure part is not in the same plane with the base, the microstructure part and the base part were observed separately from the same field of view and are named as Fig. S4 (b) and Fig. S3 (b₁) respectively. The final adhesion density was obtained by adding the two parts (for example, $N = N_b + N_{b1}$). As shown in Fig. S4 and Fig. 5a, the average spore adhesion density on unmodified “Smooth”, “Sharklet”, and p[BNIm][Br]-“Sharklet” surface are counted to be 6747 ± 535 , 5734 ± 267 , and 1728 ± 33 cells/mm², respectively for *Porphyridium*. The experimental results can also be obtained from the statistical results of *Dunaliella* (see Fig. 5b). The surface adhesion density of the previously mentioned three surfaces were 1829 ± 144 , 1367 ± 170 , and 542 ± 94 cells /mm² respectively. Although the “Sharklet” surface can reduce the adhesion density to some extent, it is far less effective than the p[BNIm][Br]-modified surface. The antifouling performance of p[BNIm][Br] is due to the alkyl imidazolium ionic liquids, which can inhibit growth and biofilm formation by permeabilizing cell membranes in microalgae [45].

To simulate the contamination of the material surface in a real marine environment and further prove the influence of surface topography and chemical composition on settlement of microalgae, a rheometer was used to create a dynamic environment. The results of adhesion on different surfaces after the external force are shown in Fig. S5 and the statistical results are shown in Fig. 5c and Fig. 5d. From the statistical results, the adhesion density of “Sharklet” surface is still lower than the “Smooth” surface. Compared with static adhesion, the two types of microalgae in the “Smooth” group were removed by 40.7 % and 36.5 % at dynamic environment (see Table S1). After modification by p[BNIm][Br], the adhesion density decreased further and the removal ratio

increased to 60.1 % and 55.9 %. It indicates that the surface adhesion of microalgae is weakened after polymer brushes modification, and it is more easily removed under external force. The reason may be that the hydrophilic p[BNIm][Br] appear in a stretched and swollen state under the action of liquid shear force, which plays a role in the release of pollutants [46]. Combining the results of static and dynamic settlement of microalgae, The poly(ionic liquid) brush-grafted “Sharklet” surface exhibits considerable antifouling performance for microalgae (*Porphyridium* and *Dunaliella*) due to the synergistic effect of the surface composition and microstructure. The “Sharklet” surface can prevent adhesion of marine microorganisms due to surface microstructure, and p[BNIm][Br] can further significantly inhibit the adhesion of pollutants owing to the presence of benzotriazole and imidazole cationic groups.

4. Conclusion and outlook

Inspired by antifouling surfaces in nature, we prepared poly(ionic liquid) brush-grafted “Sharklet” surface via 3D printing technology and subsurface initiated ROMP. The poly(ionic liquid) can graft onto the surface and subsurface of resin substrate and show obvious wear resistance. We evaluated antibacterial and anti-protein properties of poly(ionic liquid) brush-grafted hybrid materials; the p[BNIm][Br] exhibited good anti-bovine serum protein adsorption and better anti-bacterial properties against both *E. coli* and *S. aureus* due to the synergistic effect of the imidazole and benzotriazole groups. The synergistic antifouling effect of the “Sharklet” surface and the chemical composition proved to be very effective against microalgae, and the percentage of *Porphyridium* removal on p[BNIm][Br]-grafted “Sharklet” surface increased from 40.7% to 60.1%.

Author statement

Baoluo He: Investigation, Materials synthesis, Characterizations, Formal analysis, Writing-original draft, Writing-review & editing.

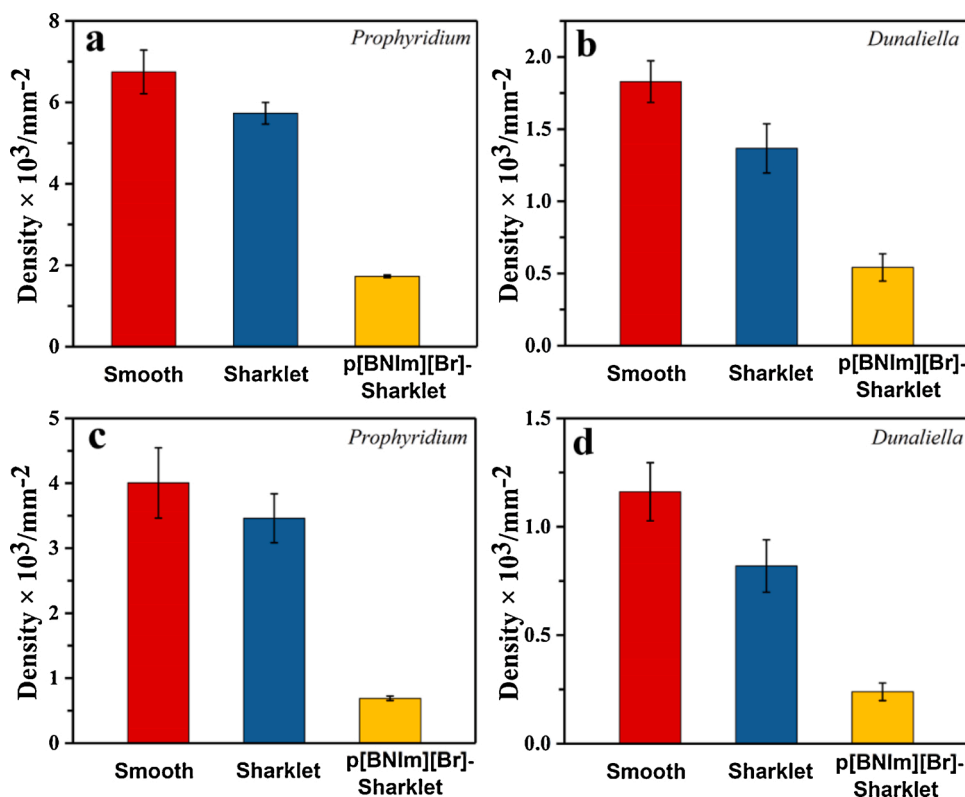


Fig. 5. Statistical results of attachment density at the static and after the external forces on different resin surfaces. (a,b) static environment; (c, d) dynamic environment.

Yixuan Du: Investigation, Data curation. Biwen Wang: Characterizations, Formal analysis. Xiaolong Wang: Review & editing. Qian Ye: Methodology, Resources, Writing-review & editing, Funding acquisition, Supervision. Shujuan Liu: Review & editing, Project administration, Supervision.

Declaration of Competing Interest

The authors report no declarations of interest.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.porgcoat.2021.106298>.

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